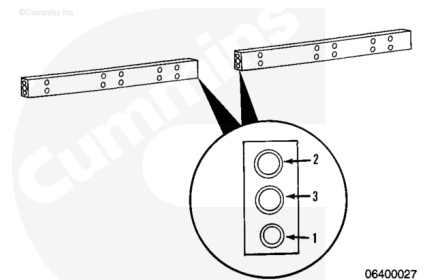


Trainings ▾

General Information

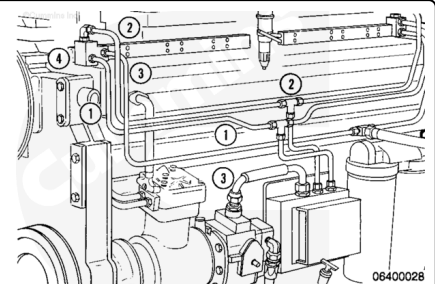
with Mechanically Actuated Injector

The QSK19 engine is equipped with a QSK fuel system that requires a split fuel manifold and three fuel circuits. The three circuits are rail (1), timing (2), and drain (3).



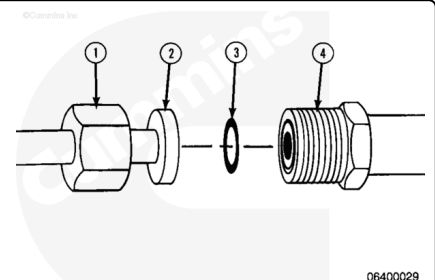
⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



Fuel is delivered to the fuel blocks (4) at the front and rear of the engine from the electronic control valve assembly.

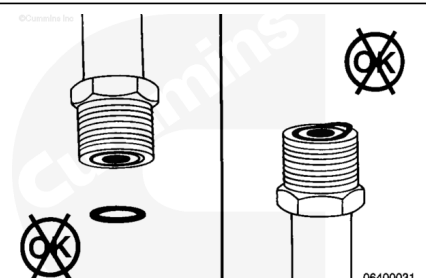
The QSK19 uses face seal o-ring fittings, except the fuel filter inlet and the fuel drain connection that are 37 degree flare fittings.



1. Tube nut
2. Tube flare (flat)
3. O-ring
4. Fitting

⚠ CAUTION ⚠

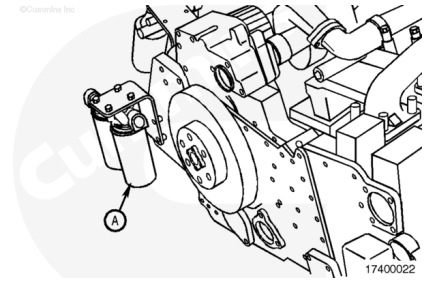
If a multi-purpose lubricant, or equivalent, is not used, the o-ring can fall out or get pinched during assembly and cause a fuel leak.



Hold the o-rings in position during assembly using Lubriplate® 105 multi-purpose lubricant, or equivalent.

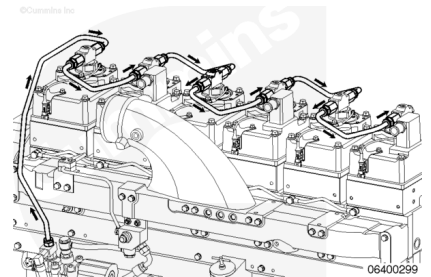
Rail Applications

The 10 micron fuel filters/water separators (A) are remote mounted on the lubricating oil pan on the front side of the engine.



with Electronically Actuated Injector

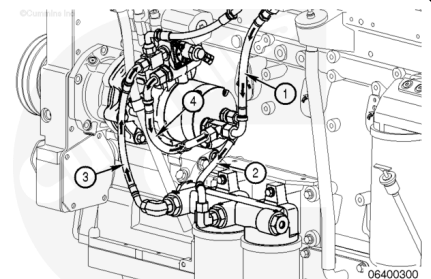
The injector supply lines carry the pressurized fuel from the fuel pump to the injectors. The injector supply lines are double wall pipe. The inner pipe carries the pressurized fuel to the injectors. The outer wall pipe is a protective shield. If there is a fuel leak, it will direct the fuel to the vent line located on the side of the fuel pump.



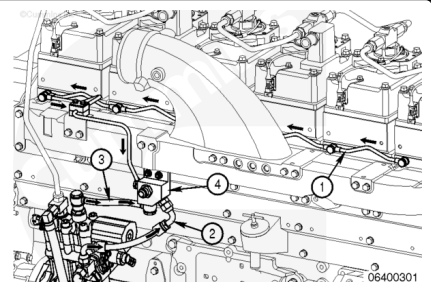
The fuel supply lines transfer fuel from the fuel inlet to the gerotor inlet, to the stage two fuel filter, and back to the fuel pump. They are flexible hoses with either banjo fittings or o-ring face seals to prevent leakage.

The fuel flow for the fuel supply lines consists of the following:

1. Fuel inlet from first stage one filters to gerotor inlet
2. From gerotor to stage two filter inlet
3. From stage two filter to fuel pump
4. Bypass from inlet actuator back to gerotor.



The fuel return lines located on the cylinder head assemblies return fuel that was not burned from the injector to the vehicle fuel tank. The fuel return line on the mechanical dump valve takes excess fuel from the fuel pump and returns it to the vehicle fuel tank. The inlet metering valve has a line which returns excess fuel to the inlet of the gerotor pump. This recirculation provides constant pressure at the inlet metering valve to prevent fuel starvation to the pump. A second line

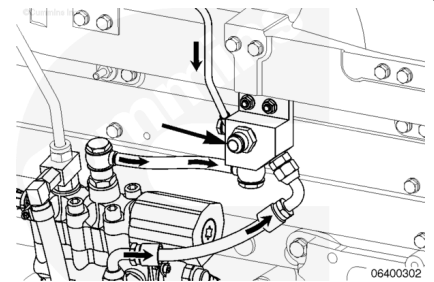


running from the inlet metering valve to the fuel return manifold is an air bleed to purge air from the fuel system (after fuel filter change).

The fuel flow for the fuel return lines consists of the following:

1. Fuel return line from cylinder heads
2. Fuel return line from inlet meter valve
3. Fuel return line from mechanical dump valve
4. Fuel manifold.

The fuel manifold provides one point where all the fuel returns connect and the fuel is returned to the vehicle fuel tank. The fuel manifold has an air bleed fitting for the fuel return line from the inlet metering valve. This air bleed fitting is a check valve that opens at 5 bar [72 psi]. This limits flow rates out of the system at higher pressures and minimizes any flow loss during cranking and startup.



The injectors are made of three major components:

- A tee-connector which provides threaded ports for the injector supply lines to connect to.
- A small screen to collect debris, an orifice, and a flow limiter to help maintain a constant supply pressure in the injector.
- An injector body provides space to be an accumulator for the pressurized fuel. An electrical lead with connector is plugged into the main engine harness. An electronically controlled solenoid is used to operate the needle. The ECM commands the solenoid to control fuel injection and control when combustion occurs.

